

On the minimum dataset size and quality for a maize seedling object detector

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Keywords: keyword 1; keyword 2; keyword 3 (List three to ten pertinent keywords specific to the article; yet reasonably common within the subject discipline.)

1. Introduction

1.1. The Problem of Plant Counting

Plant counting is a critical operation in precision agriculture, plant breeding, and agronomical evaluation. Accurate plant counts can provide valuable information for both farmers and researchers. This task was often performed manually by human operators. Today, this process can be automated by the use of computer vision algorithms. To validate a method for counting, it is critical to set a benchmark for accuracy. A benchmark can be defined by the accuracy of manual counting, international standards, or by comparison with other already accepted methods. Accuracy of plant manual counting depends on human performance, so variable, but often taken as golden sample. According to the European Plant Protection Organization, the benchmark for acceptance is a coefficient of determination (r^2) of 0.85 when compared to manual counting [1]. This corresponds to a Root Mean Square Error (RMSE) of approximately 0.39. Also scientific literature mention a r^2 value of 0.85 with ground truth (manual counting) as a benchmark for acceptance [2].

1.2. Case study: Maize Seedling Counting

Literature shows that benchmark for acceptance can be achieved with computer vision object detection [3–5] or regression models [6,7]. A superiority of object detection over regression models in terms of accuracy was found [2]. Object detection is also more versatile than regression models, because, object detection model inference on georeferenced

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orthomosaics can be used to verify plant presence through its coordinates throughout the vegetative phases. This is not possible with regression models, because they can only predict the number of plants in an image. The ability to identify and locate the objects within an image are the main features of object detection models. The validation of this ability rely on metrics as Intersection over Union (IoU), Average Precision (AP), and Average Recall (AR) [8] rather than the coefficient of determination. Identification supports are usually bounding boxes, which are rectangles that enclose the object of interest, but they can also be points or other kind of geometries. Counting of plants is then commonly performed by counting the number of geometries that enclose the objects of interest (the plant) in a image area. Georeferenced orthomosaics are images created through aerial photogrammetry, a process that involves capturing overlapping georeferenced images taken by nadiral view picturing georeferenced ground control points, and performing boundle adjustment to form a single, seamless image [9]. Georeferenced orthomosaics are scaled and oriented to a geographical coordinate system. It implies that the pixel coordinates corresponds to geographical coordinates and are projectables in a metric system. Using georeferenced orthomosaics in seedling counting makes object detection easier, because the metric scale can be used to locate the objects in the image and prospective variance is reduced to zero.

Plant counting is affected, as many object detection application fields, from data scarcity: public datasets are rare and often not suitable for the task because of the large environmental variability and their images are not orthorectified or scale is unknown. Even so we can find some datasets that can be used for training a plant counting object detector [10]. Select a specific crop at a specific growth stage can reduce the variability that the model should learn and make it possible to better study the other variables that can affect the dataset size and quality requirements. For this study we choose grain maize seedlings (*Zea mais* L.) at the V3-V5 (BBCH 13-15) growth stage [11], because it is the most represented plant in scientific [3,4] and not scientific [12,13] open datasets from aerial photogrammetry. Maize is a commodity crop that is widely grown in the world and is the most important crop in the world by production [14]. Grain maize seedlings in that stage are easy to count because of their low overlapping and fixed intra-row and inter-row spacing, differently by silo-mais seedling that is seeded with extremely low inter-row spacing. This particular outfit that makes grain maize seedlings a good candidate for object detection, is shared with other crops that are seeded in rows with inter-row spacing such as sunflower (*Helianthus annuus* L.) or sugarbeet (*Beta vulgaris* L.).

1.3. Object Detection approaches

The development of object detection algorithms has evolved from not machine learning methods, here named handcrafted methods (HC), to modern deep learning-based (DL) techniques. Today, all the state-of-the-art object detection methods are based on DL models. Nethertheless, HC methods are still used in some cases [4,15]. Most of modern DL object detection uses convolutional neural networks (CNN) [16] based frameworks (e.g., Faster R-CNN [17], YOLO [18]) or Transformer [19] based approaches (e.g., DETR [20]) or mixuturs of both approaches. The main difference between CNN-based and Transformer-based models is the way they process the image. CNN-based models process the image in a grid-like fashion (convolutions), while Transformer-based models process the image as a sequence of patches (attention mechanisms) [21]. On common benchmarks as COCO [8], PASCAL VOC [22], and ImageNet [23], Transformer-based models have shown to be more accurate than CNN-based models [24]. One of the critical challenges in training effective DL object detection models is the availability of a sufficiently large and high-quality dataset. CNN-based models are still widely used in object detection, because they are more efficient in processing small images and have a lower computational cost [25]. However, when

fine-tuning with scarce data, Transformer-based object detectors generally perform better than CNN-based detectors, provided they are pretrained on large datasets [26–28].

To represent the categories and compare performance between pure-CNN and Transformer-mixed architectures that are effectively used as object detectors in real plant counting applications, YOLOv5 and YOLOv8 can be taken as pure-CNN architecture representatives for their large use in agriculture [29]. Their large diffusion in agriculture applications is justified by the fact that they leverage good precision and the low need in terms of dataset size in respect other CNN architectures [30–32]. YOLOv5 [33] is a pure CNN-based architecture that represents a significant optimization over previous YOLO versions with improved speed-accuracy tradeoffs and enhanced training stability. It employs a CSPDarknet53 [34] backbone that uses cross-stage partial connections to reduce computational costs while maintaining feature representation power. YOLOv8 [33] introduced architectural modifications including a decoupled detection head, a C2f module for feature processing, and several advanced augmentation techniques, resulting in better performance in respect to YOLOv5. Its backbone is an improved CSPDarknet with enhanced SPPF (Spatial Pyramid Pooling - Fast) modules for multi-scale feature extraction. Real-Time-DETR (RT-DETR) is a recent Transformer-mixed architecture that outperforms YOLOv5 and YOLOv8 [35]. RT-DETR [35] is a real-time transformer-based detector that combines CNN backbone features (the ResNet-based CSPResNet50 and CSPResNet101 backbones [36]) with a streamlined transformer decoder. YOLO11 has been recently proposed as a Transformer-mixed YOLO architecture that outperforms RT-DETR [37]. YOLO11 [37] introduced several architectural innovations including adaptable block and module designs, adaptive head assignments, and enhanced loss functions for optimizing detection at various scales. YOLO11 uses an advanced adaptable backbone with improved feature pyramid networks for efficient multi-scale detection.

Recently, zero-shot and few-shot object detection have emerged as promising paradigms to alleviate the need for large annotated datasets. While traditional object detection models (many-shots object detectors) require extensive labeled data for training, few-shot and zero-shot object detection aim to detect novel objects with little to no labeled examples respectively. The term "shot" refers to the number of annotations used to train the model over all the images. Each shot corresponds to an individual, that in the case of few-shots, is used to prototype the object of interest. Zero and few-shots approaches often leverage feature transfer or meta-learning components to generalize across classes under extreme data scarcity [38]. This approach reduces the annotation burden and is especially beneficial in domains where collecting exhaustive training data is impractical. Zero-shot object detection detects new categories without any training samples by leveraging semantic relationships or contextual information learned from known classes [39]. Few-shot approaches optimize models to learn quickly from a handful of labeled examples [40]. Meta learning is a common approach in few-shot object detection [41–44], while Cross-Domain Few-Shot Object Detection (CD-FSOD) has recently surpassed this approach by leveraging domain adaptation techniques [45]. DE-ViT [46] and CD-ViTO [47] are the latest models that have shown promising results in few-shot object detection. CD-ViTO is a few-shot object detection model that specifically addresses the challenges of domain shift between base and novel classes. The backbone consists of a modified Vision Transformer (ViT) [21] structure with domain-invariant attention layers that help extract features robust to domain variations. CD-ViTO builds upon an open-set detector framework by incorporating several key modules to mitigate domain discrepancies. In particular, it enhances the baseline model by making the instance features learnable (MLIF), which adapts fixed, pretrained features to better capture the target semantics. This is complemented by an instance reweighting module (MIR) that assigns higher importance to high-quality samples—those with clear,

unambiguous boundaries—thereby forming more robust class prototypes. Furthermore, a domain prompter (MDP) is introduced to simulate virtual domain shifts, forcing the network to produce consistent representations despite changes in visual styles. Finally, a lightweight fine-tuning step is applied to critical detector components, further refining the model's ability to generalize with minimal annotated examples. Together, these innovations enable CD-ViTO to effectively bridge the gap between base and novel domains, achieving state-of-the-art performance in few-shot object detection tasks across varied target datasets. Zero-shot actual state-of-the-art object detector models are YOLO-World [48], OWLv2 [49], and Grounding DINO [50]. OWLv2 (Open-World Localization v2) is an zero-shot object detector that represented a significant evolution in open-vocabulary detection capabilities. It employs a vision-language foundation model approach that enables detecting objects described by arbitrary text prompts without specific training data for those categories. The model uses a dual-encoder architecture with a vision encoder (based on a scalable ViT backbone) and a text encoder (based on BERT [51]), connected through contrastive learning objectives. OWLv2's design includes multi-scale feature fusion, region-text contrastive alignment, and localization refinement modules that allow it to accurately detect objects across various sizes and contexts. Unlike traditional detectors requiring class-specific training, OWLv2 leverages its pre-trained knowledge of visual concepts and their textual descriptions to generalize to novel object classes.

1.4. Dataset Size and Quality Requirements

Many studies have been conducted on object detection for plants in open field [3,5,6,52–62], but few have focused on dataset minimum requirements for training a robust plant object detector [4,63]. Some study focused on few-shots approach for plant counting [61,64], but only two specifically accounting on few-shots method for maize seedling counting [65,66]. Unfortunately, the lonely two studies evaluating few-shot performance on maize seedling counting by orthomosaics do not achieve the benchmark and do not clearly specify the number of shots used. Also no zero-shot benchmark for maize seedling counting has been set yet, and no research on this application has been done yet.

Even if no research has been made in order to minimize or set a benchmark for the dataset size and quality required to train a maize seedling object detector, it is well known that the performance of a DL model is directly related to the amount of data used for training [67] and its quality [68]. Dataset size and quality predictability in DL has been proven to be addressable with empirical approaches [69,70]. As already mentioned, model architecture is critical in dataset requirements as different models may require different dataset sizes and qualities to achieve the same performance [71,72]. Backbone is pivotal on downstream tasks as object detection [73]. The importance of using a domain specific backbone has been proven also for plant/leaves segmentation on orthomosaics [74], but backbone training is still prohibitive for many applications. No backbone weights specialized on agricultural orthomosaics has been published yet, as for the most of the specialized domains. Even if a so specialized backbone exists, some concern will come out about its out-of-distribution generalization capability [75]. So, because of limited resources, today only the use of a general backbone is possible in practical sense [76], even if a decreasing in dataset size requirements is expected with a domain specific backbone because of the out-of-distribution generalization [77]. Another factor that can affect the dataset size and quality requirements other than the already cited model architecture and use of pre-trained backbones, is the training data augmentation strategy. Some studies proposed highly computationally expensive data augmentation such as trainable data augmentation [78] or data augmentation with generative adversarial networks [79]. Other studies proposed to use less computationally expensive data augmentation strategies [80,81] and someone even

proved that random image augmentation can provide equivalent results to more expensive techniques [82].

Nethertheless, the most important factor affecting DL training is the dataset himself [67] and its source. It has been observed that using training samples from the same dataset as the inference (in-domain dataset) dramatically increases accuracy and reduce the need for a large dataset in respect to collecting training samples from other datasets (out-of-distribution dataset) [4,63]. From comparison of the studies here mentioned, dataset source seems to be the most important factor in determining the dataset size and quality needed to achieve the benchmark for acceptance.

1.5. Study Aim

This study aims to determine the minimum dataset size and quality required to train the benchmark reaching object detection model for identifying maize seedlings in georeferenced orthomosaics, where the dataset size and quality are respectively defined by the amount of annotated images in the training set, and the accuracy of the annotations. Also the effect of training dataset source will be evaluated in this study. Models architecture and size will be taken into account, as long as the object detection downstream task is concerned. After setting the dataset size and quality minimum requirements for many-shots object detectors, we will evaluate if new zero-shot and few-shot object detection models can achieve the same benchmark for acceptance with less data and annotations. We will also discuss the need for HC method in a DL object detection pipeline.

2. Materials and Methods

2.1. Datasets

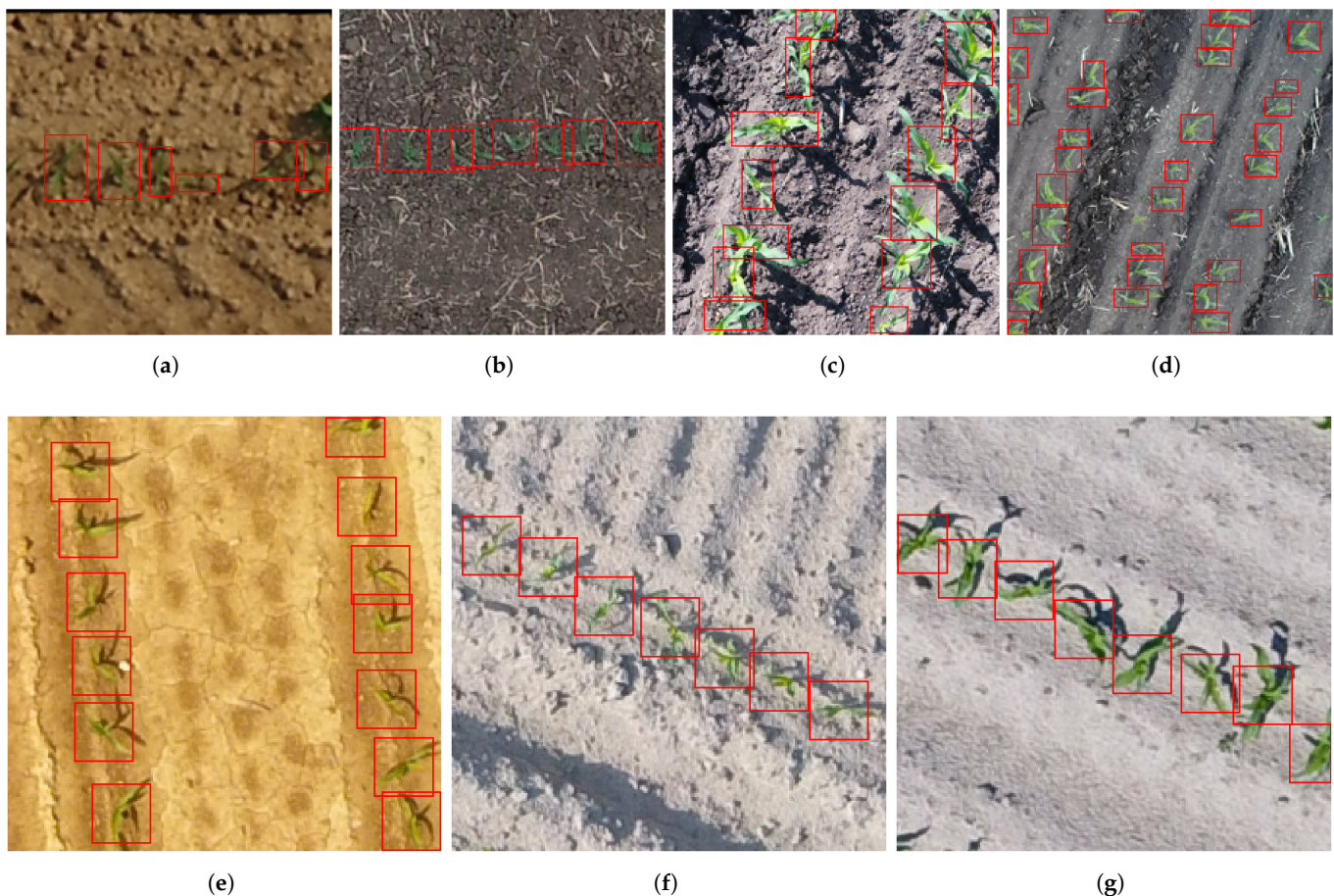
The datasets used in this study to train the object detection models for maize seedlings counting are nadiral or supposedly nadiral images of maize seedlings at the V3-V5 growth stage or estimated so. The V3-V5 growth stage is defined by the BBCH scale as the stage where the third to fifth leaf is unfolded and the plant is 15-30 cm tall [11].

This study uses two dataset sources as training sets: the out-of-distribution dataset (OOD) and the in-domain dataset (ID). The ID datasets are from the same source as the testing dataset, while the OOD datasets are not. The OOD datasets are composed of images from scientific literature [4,6] and from internet repositories [12,13]. The ID datasets were collected during this study. This ID dataset creation consisted in capturing nadiral images of three study areas with a Phantom 4 Pro v2.0 (DJI, Shenzhen, China) drone equipped with its default series RGB camera @ 10 m AGL (above the ground level). The number of images captured depends on the study area size that was about 2 hectares for ID_1 location and about 1 ha for the other two. For each location, an orthomosaic was created using a photogrammetric software. Bundle adjustment error was estimated as 38 mm using the ground control points surveyed by GNSS operating in VRS-NRTK mode. The orthomosaics were generated with an average ground sampling distance of 5 mm/pixel in the WGS84/UTM 32 N reference system.

The OOD scientific datasets and the ID datasets are composed by patches of georeferenced orthomosaics of known scale, while the OOD internet datasets are simple RGB images of unknown scale. The all OOD datasets came with annotations, while the ID datasets were manually annotated.

Table 1. Summary of Datasets Used in the Study

Dataset	Description
OOD Scientific	Composed of tiles of georeferenced orthomosaics of maize seedlings from scientific literature. - DavidEtAl.2021 [4] 182 tiles - LiuEtAl.2022 [6] 596 tiles
OOD Internet	Composed of RGB images of maize seedlings from internet repositories. - Maize stage V3 [12] 216 tiles - Maize stage V5 [13] 174 tiles
ID	Collected during this study, composed of tiles of georeferenced orthomosaics of maize seedlings of known scale. - ID_1 Maize stage V3, tiles: 150 training, 20 testing - ID_2 Maize stage V3, tiles: 150 training, 20 testing - ID_3 Maize stage V5, tiles: 150 training, 20 testing

**Figure 1.** (a) DavidEtAl.2021, (b) LiuEtAl.2022, (c) Internet Maize stage V3, (d) Internet Maize stage V5, (e) ID_1, (f) ID_2, (g) ID_3.

To make the two kind of dataset comparables we chose to rescale the images to a scale of 0.005 m/pixel where the scale was known (scientific OOD and ID datasets), obtaining orthomosaics of different sizes. All the orthomosaics were then cropped to 224*224 pixels tiles. Also the annotations were rescaled and cropped where needed. Figure 1 shows a sample for each dataset. Each ID dataset has 20 tiles to be used as testing dataset, while other 150 tiles are used as training dataset.

2.2. Handcrafted object detector

Like other works [4,6,15] we wrote an HC algorithm to get annotated tiles from the orthomosaics, basing it on agronomical knowledge and color thresholding. Hue, saturation and value (HSV) color space was used here to threshold the image, to get green pixels, but other color spaces can be used. For the execution of this algorithm, the following graphical and agronomical parameters must be set: color minimum and maximum thresholds (color threshold), the minimum and maximum leaf area for plant (leaf area range), the minimum distance between plants on rows (intra-row distance), and the distance between rows (inter-rows distance). The algorithm is expected to work: on orthomosaics of maize seedlings at the V3-V5 growth stage, with low weeds infestation, with rows having roughly the same angle with meridian and distance between them. The algorithm is divided in two parts. The first part, named HC1 Algorithm A1, that take as input the orthomosaic and gives back a list of region polygons that enclose the supposed plants. HC1 is expected to give a poor accuracy plant count on all the processed tiles. Thus we implemented a second process named HC2 Algorithm A2 that is intended to filter in only the tiles with an accurate plant count. Algorithm pseudocode is provided in appendix.

2.3. Deep Learning object detectors

The many-shots object detectors used in this study are YOLOv5, YOLOv8, RT-DETR and YOLO11. We chose them for the considerations made in Introduction Section 1.3. We used here the Ultralytics implementation of these models because the implementation is open-source [83], and it makes it possible to tune parameters size coherently across all tested architectures.

For all the many-shots models we used the same hyperparameters and augmentations as the library default, with the following exceptions:

- batch size: 16
- maximum training epochs: 200
- maximum training epochs without improvement: 15
- precision: mixed

All the many-shots models were trained with a training dataset split of 90% training and 10% validation. To avoid miss-predictions due to object on the border of the tile, the predicting image was extended by a buffer of half the size of the tile around the tile. The trained models were tested on the testing dataset tiles with the SAHI method [84]. The SAHI method is a method that slices the testing image in patches of the same size of the training tiles and then tests the model on each of them. The model outputs of each patch are then merged by non-maximum suppression and cropped by the original tile extension.

To test the few-shot approach we trained CD-ViTO with multiple model sizes. The size of this model is determined by the backbone used, which can be ViT-S, ViT-B, or ViT-L [85]. We used the implementation of CD-ViTO provided by the authors [47]. In the context of this study a 'shot' correspond to an image with a single annotated plant. We used 1, 5, 10, 30, and 50 shots to train the model. The shots were randomly selected from the ID manually labeled dataset, then a random annotation was selected from the same image to be used as prototype.

For testing the zero-shot approach we used OWLv2. We took this architecture as zero-shot object detector example as it is the most stable state-of-the-art model for this task [49,50]. For test OWLv2 we used the implementation of the transformer library [86] with the parameters published by the authors. We tested the encoder sizes ViT-B/16, ViT-L/14 with the following three pre-training strategies:

- **Base models:** Trained using self-supervised learning with the OWL-ST method, which generates pseudo-box annotations from web-scale image-text datasets
- **Fine-tuned models:** Further trained on human-annotated object detection datasets
- **Ensemble models:** Combining multiple weight-trained versions to balance open-vocabulary generalization and task-specific performance

For all the OWLv2 variants, we tested multiple text prompts to describe maize seedlings:

- "maize"
- "seedling"
- "plant"
- "aerial view of maize seedlings"
- "corn seedlings in rows"
- "young maize plants from above"
- "crop rows with corn seedlings"
- "maize seedlings with regular spacing"
- "top-down view of corn plants"
- "agricultural field with maize seedlings"
- "orthomosaic of corn plants in rows"

The best performing prompt was selected for each model variant based on validation performance.

Table 2 shows the architectures used in the study with the parameter size specifics.

Table 2. Summary of Tested Architectures and Model Sizes¹

Architecture	Shots	n	s or S	m or B	l or L	e
YOLOv5	many	1.9	7.2	21.2	46.5	86.7
YOLOv8	many	3.2	11.2	25.9	43.7	68.2
YOLO11	many	4.0	12.5	28.0	50.0	75.0
RT-DETR	many	-	-	-	60.0	80.0
CD-VITO	few	-	22.0 ²	86.0 ³	307.0 ⁴	-
OWLv2	zero	-	-	86.0 ⁵	307.0 ⁴	-

¹ Values represent millions of parameters or used backbone.

² ViT-S ³ ViT-B ⁴ ViT-L

2.4. Minimum dataset size and quality modelling

In order to investigate the minimum size and quality of the dataset required to train a robust object detection model for maize seedlings counting, we conducted a series of experiments where the above mentioned DL models were recursively fitted with increasing dataset size and quality. For many-shots models we consider a training dataset split of 10% validation and 90% training, while for few-shots the number of shots determined the amount of training samples. Zero-shots relied only on descriptions in natural language of the objects to be detected. For what concerns only the dataset size evaluation, for many-shots models we considered sizes from 10 to 150 images in 15 steps of 10 images, while for few-shots models we considered 1, 5, 10, 30, and 50 shots. For what concerns the dataset quality, we evaluated the annotation quality by

For all the models we evaluated the relationship between dataset size or quality and model performance using R-squared and mAP, respectively for plant counting and plant detection. Whether R-squared provided values below -1 we also considered RMSE as metric for counting. MAPE was considered for few-shots and zero-shots models only to evaluate the quality of the annotations produced by the prediction of these models. We list here the metrics formulas for clarity:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (1)$$

where y_i is the ground truth count for the i -th image, $\hat{y}_i$ is the predicted count, and $\bar{y}$ is the mean of all ground truth counts. R^2 ranges from $-\infty$ to 1, with 1 indicating perfect prediction, 0 indicating that the model predictions are no better than simply predicting the mean, and negative values indicating that the model performs worse than predicting the mean.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (2)$$

where RMSE measures the average magnitude of prediction errors in the original units (number of plants).

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (3)$$

where MAPE measures the percentage error relative to the actual values, providing a scale-independent measure of accuracy. It is expressed as a percentage, with lower values indicating lower percentage of false positive or false negative. Thus it was reported as an index of the quality of the annotations. Note that MAPE is only calculated for cases where $y_i \neq 0$ to avoid division by zero. It is particular useful for counting as testing tiles never have zero plants.

For object detection performance, we used the standard COCO evaluation metric:

$$\text{mAP} = \frac{1}{|\text{IoU}|} \sum_{t \in \text{IoU}} \text{AP}_t \quad (4)$$

where mAP (mean Average Precision) is calculated at a single IoU (Intersection over Union) threshold of 0.5. AP at the IoU threshold is the area under the precision-recall curve for detections that meet that IoU threshold criterion.

To test the predictability minimum dataset size and quality required to train a robust (achieving benchmark) object detector for maize seedlings counting through empirical models, we test the logarithmic, arctan and algebraic root functions to fit the dataset size or quality versus performance relationships as suggested by previous studies [70]. For clarity we list here the functions tested:

$$\text{Logarithmic: } f(x) = a \ln(x) + b \quad (5)$$

$$\text{Arctan: } f(x) = a \arctan(bx) + c \quad (6)$$

$$\text{Algebraic Root: } f(x) = ax^{1/b} + c \quad (7)$$

For the model fits to dataset size versus performance relationships, we evaluated multiple fitting functions and selected the one with the highest goodness-of-fit:

$$\text{GoF} = R_{\text{fit}}^2 = 1 - \frac{\sum_{i=1}^n (y_i - f(x_i))^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (8)$$

where y_i is the observed metric (either R-squared or mAP), $f(x_i)$ is the fitted value at dataset size x_i , and $\bar{y}$ is the mean of the observed metrics.

3. Results

3.1. Handcrafted object detector

To evaluate the HC object detector as a training dataset extractor from the ID dataset, we measure amount of annotated tiles that the HC algorithm can extract from the orthomosaics and the accuracy it deliver in compare to handmade annotations. The table 3 shows the performance of the HC object detector on the ID datasets by enumerating metrics and succesfully annotated tiles. The metrics were computed on the testing dataset tiles.

Table 3. HC Object Detector Performance

Dataset	R-squared	RMSE	MAPE	mAP	tiles	dataset %
ID_1	0.95	0.12	9%	0.87	1184	7.8%
ID_2	0.93	0.11	12%	0.81	279	4.2%
ID_3	0.87	0.18	16%	0.73	158	1.8%

3.2. many-shots object detectors

The models were evaluated with the following metrics: R-squared, RMSE, IoU, AP, and AR. All the metrics were computed for eight different score thresholds for the same model to evaluate the model performance at different confidence levels. The values to thresholds bounding boxes score were: 0, 0.29, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 0.95. The highest R-squared value within the eighth probability thresholded was considered as the model performance for that experiment.

3.2.1. OOD training

The OOD scientific datasets "DavidEtAl.2021" and "LiuEtAl.2022" were tested singularly and in combination in the experiment named "scientific OOD". The OOD internet datasets "internet OOD" was tested singularly and in combination with the OOD scientific datasets in the experiment named "All OOD". No one of the dataset combinations reached the benchmark R-squared value of 0.85 with the any model. The coefficients of determinations and the root mean square errors for all the OOD experiments count tests are shown in 2.

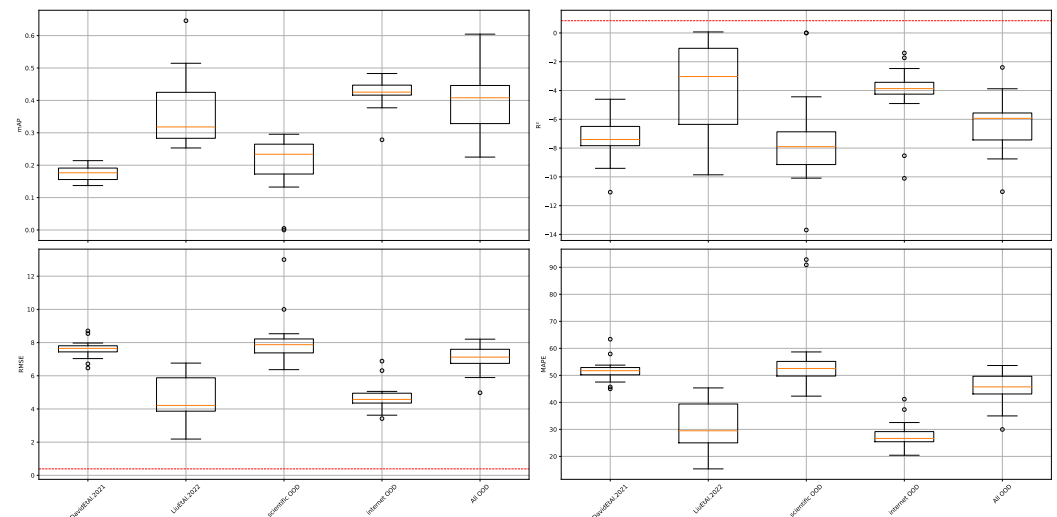


Figure 2. R-squared and RMSE values versus dataset source for all model variants. Red dashed line on r-squared indicates the benchmark threshold of 0.85.

As no model achieved the benchmark no study was done on the dataset quality requirements.

3.2.2. ID training

The relationship between ID training dataset size and model performance was evaluated for all model architectures and sizes as shown in Figure 3. The three performances of each model for a given dataset size correspond to the test on the three ID datasets. For some test, the resulting R-squared value of counting was below 0 (data not shown). Where it was not the case, it has been possible to calculate the mean and the interval of the R-squared values. It has been always possible for mAP. The R-squared values of the counting and the mAP values for all models were regressed against the dataset size using a logarithmic, root or arctan model. The goodness-of-fit (GoF), estimated as the coefficient of determination of these models, was also calculated. A high GoF value indicates that model performance is highly predictable by dataset size. Conversely, a poor GoF could indicate that other variables play a more important role in determining model performance, or that the chosen dataset size interval is too narrow to achieve a good fit.

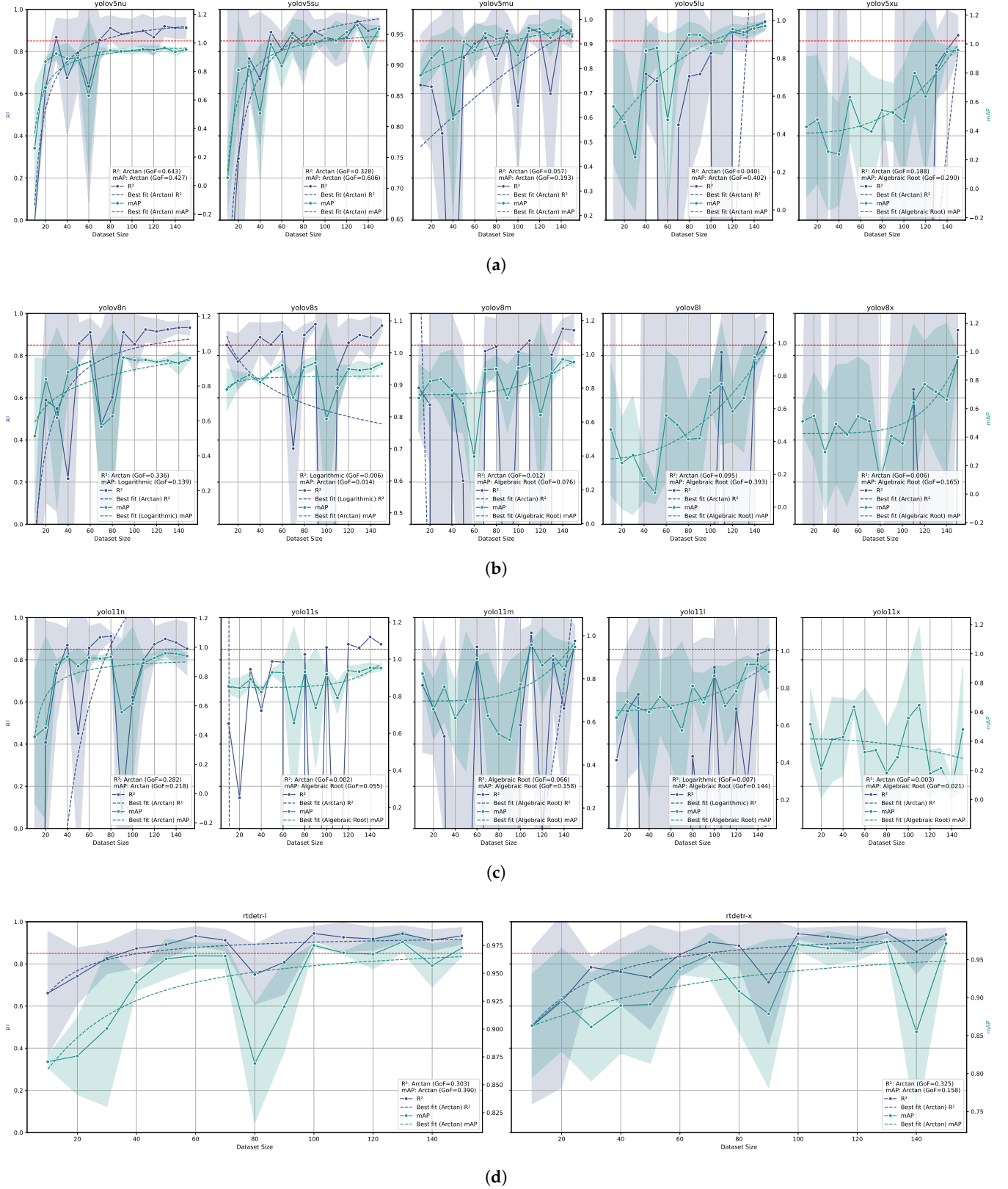


Figure 3. Relationship between dataset size and model performance for different object detection models (YOLOv5 (a), YOLOv8 (b), YOLO11 (c), and RT-DETR (d)) trained and tested on ID datasets. On the same line, each subplot represents a different parameters size of the model, increasing from the left to the right. The x-axis represents the dataset size, while the left and right y-axis represents the R-squared and mAP values respectively. The solid lines represent the mean values, while the dashed lines indicate the logarithmic fit. The shaded area around the solid lines represents the confidence interval (standard deviation) of R-squared or mAP. The red dashed horizontal line represents the benchmark R-squared value of 0.85. The combined legend in the lower right corner of each subplot shows the goodness of fit (GoF) for both R-squared and mAP.

For the combinations model-architecture/dataset-size that achieved the benchmark, the minimum dataset quality required to achieve the benchmark was evaluated as shown in Figure 4.

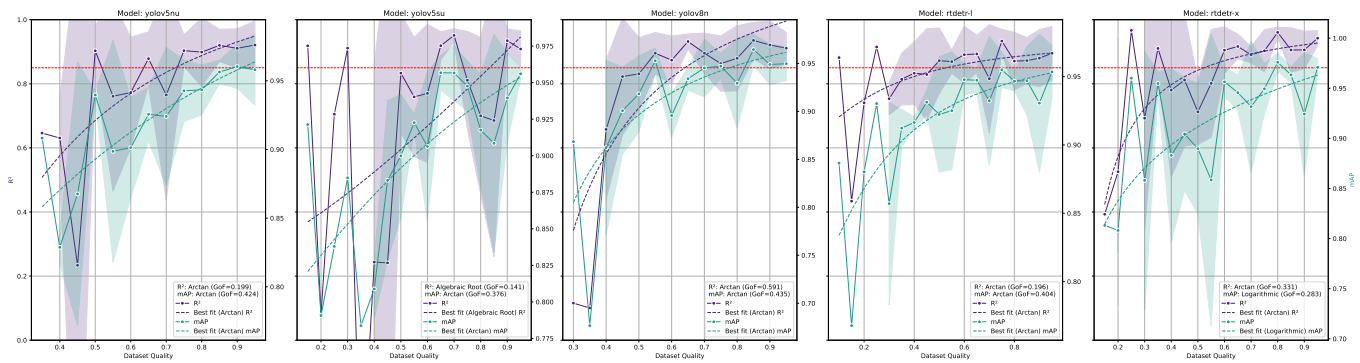


Figure 4. Relationship between dataset quality and model performance for all object detection models that achieved the benchmark. The x-axis represents the dataset quality, while the left y-axis represents the R-squared values. The red dashed horizontal line represents the benchmark R-squared value of 0.85. The legend in the lower right corner of the subplot shows the goodness of fit (GoF) for R-squared.

Within YOLO models YOLOv5n, YOLOv5s and YOLOv8n achieve the benchmark R-squared value of 0.85 with 130, 130 and 110 samples, respectively, if considered the dataset sizes where all the three model performances were above 0.85 r-squared and the logarithmic model predicted over-benchmark values for that dataset size. RT-DETR l and RT-DETR x achieve the benchmark R-squared value of 0.85 with 60 and 100 samples respectively, with the same assumptions as for YOLO models. For these models the GoF was above 0.3, while for the models that did not reach the benchmark R-squared value the GoF was always below this value. The mAP seems to follow the same trend as the R-squared values. All the models show a clear trend of increasing R-squared and mAP values as the dataset size increases, as expected. It is also clear that increasing number of parameters and model complexity for mostly CNN like models (YOLOs) leads to increasing need for dataset size. For the mostly transformer like models (RT-DETRs) it is not that clear, also because of the low amount of model parameter size tested. The confidence interval reduction in function of the dataset size indicate that variability in performance decreases significantly as dataset size increases for all models. Taking into account the dataset quality in the same way as done for the dataset size, both quality tests and quality models achieved the benchmark with the 85%, 90%, 85% and 65% of the original dataset quality for YOLOv5n, YOLOv5s, YOLOv8n and RT-DETR x, respectively. RT-DETR l did not achieve the benchmark for any dataset quality reduction tested.

3.3. few-shots object detectors

The few-shots models were evaluated with the following metrics: R-squared and RMSE because no one of the models reached the benchmark.

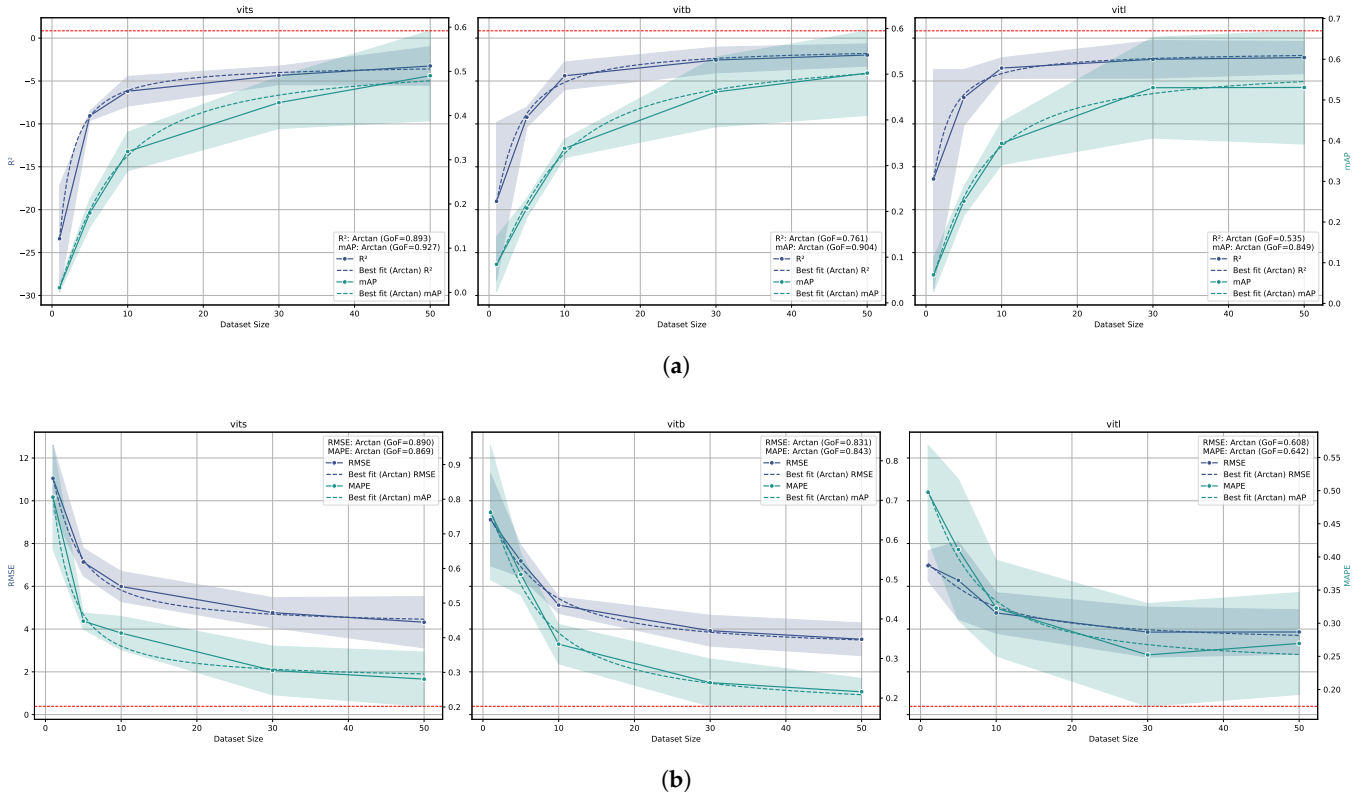


Figure 5. The figure shows the relationship between shots and model performance for the CD-ViTO model trained and tested on ID datasets. The x-axis represents the number of shots. The solid lines represent the mean values, while the dashed lines indicate the shots amount/metric prediction model. The shaded area around the solid lines represents the confidence interval (standard deviation) of the metric. **(a)** The left and right y-axis represents the R-squared and mAP values respectively. The red dashed horizontal line represents the benchmark R-squared value of 0.85. The combined legend in the lower right corner of each subplot shows the goodness of fit (GoF) for both R-squared and mAP. **(b)** The left and right y-axis represents the RMSE and MAPE values respectively. The red dashed horizontal line represents the benchmark RMSE value of 0.39. The combined legend in the upper right corner of each subplot shows the goodness of fit (GoF) for both RMSE and MAPE.

The best result achieved by the CD-ViTO model was a RMSE of 3.9 with ViT-B backbone and 50 shots to build the prototypes. It corresponds to a MAPE on counting of about 25% and a mAP of about 0.5 (Figure 5). Some of the best predictions of this model are shown in Figure 6. The models fitted on metrics show a reliable GoF for all the metrics, indicating that the model performance is highly predictable by the number of shots. These also show that any CD-ViTO size model would not achieve the benchmark with any shot amount.

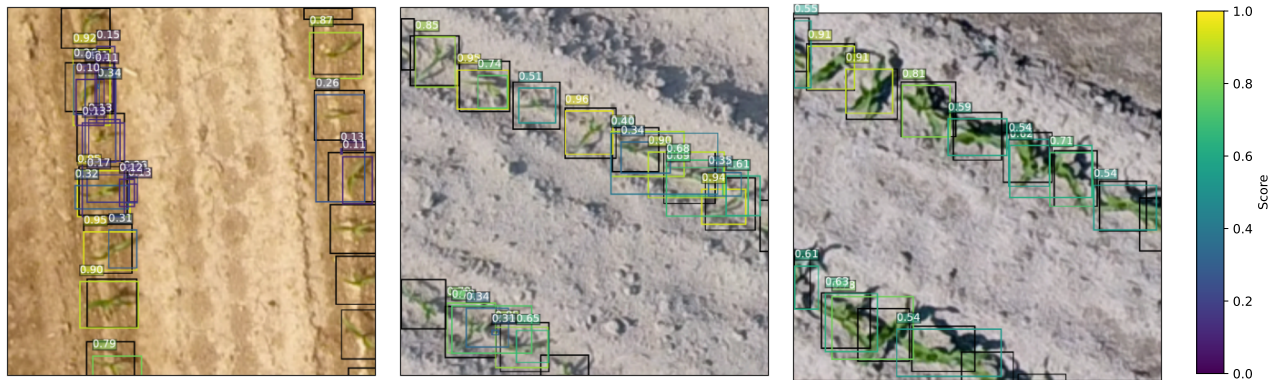


Figure 6. 50 shots CD-ViTO with ViT-B backbone predictions on the 1, 2 and 3 ID test datasets tile examples, respectively from the left hand side to the right. Black bounding boxes are the ground truth annotations, while the bounding boxes in viridis color scale are the model predictions.

3.4. zero-shots object detectors

Figure 7 shows the relationship between the OWLv2 model settings and model performance for the OWLv2 model tested on ID testing datasets. A real dataset size relation with metrics could not be established as zero-shots models do not require fine-tuning training data. No one of the zero-shots model settings reached the benchmark. This is particularly true for the R-squared values that were always below 0. The RMSE and MAPE were higher than the observed in the few-shots models. The mAP values were always above 0.5, but never reached the benchmark.

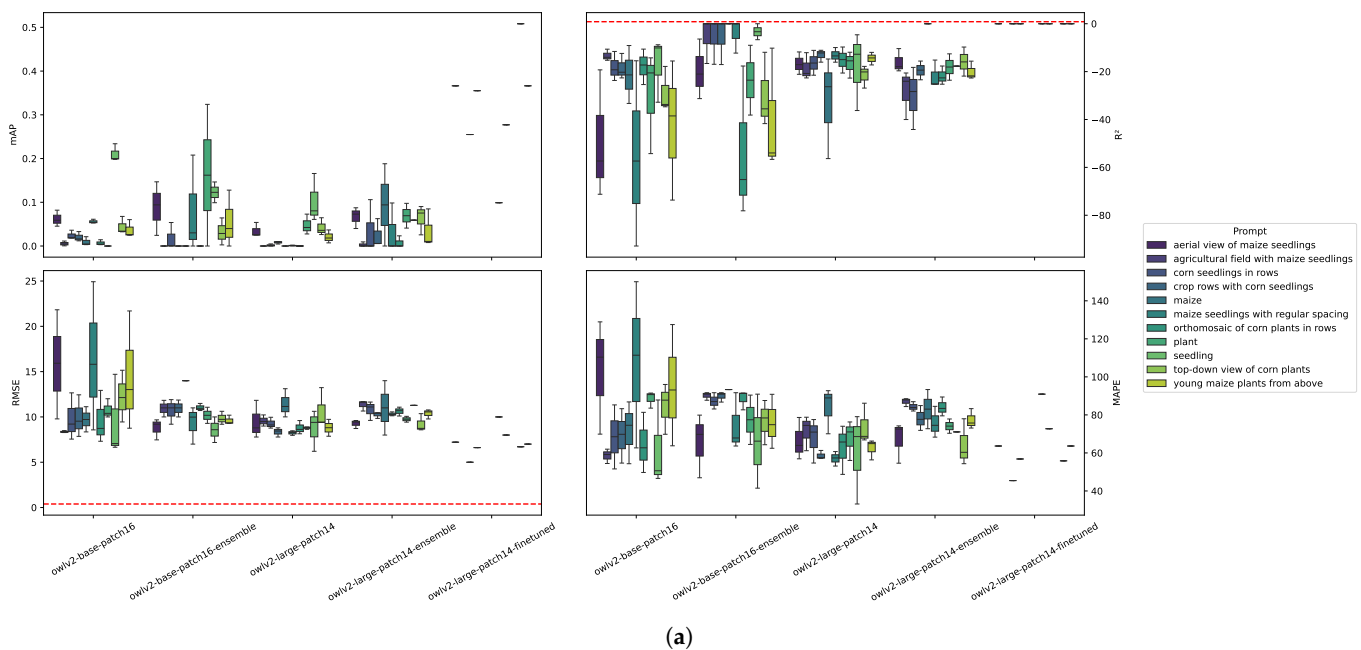


Figure 7. The figure shows the relationship between the OWLv2 model size, used prompt and model performance. The x-axis represents the model settings and the model size. Colors represent the different prompts used. The four subplots show the mAP (upper left corner), R-squared (upper right corner), RMSE (lower left corner), and MAPE (lower right corner) values. The red dashed horizontal line in the R-squared and the RMSE subplots represents respectively the benchmark of 0.85 and 0.39.

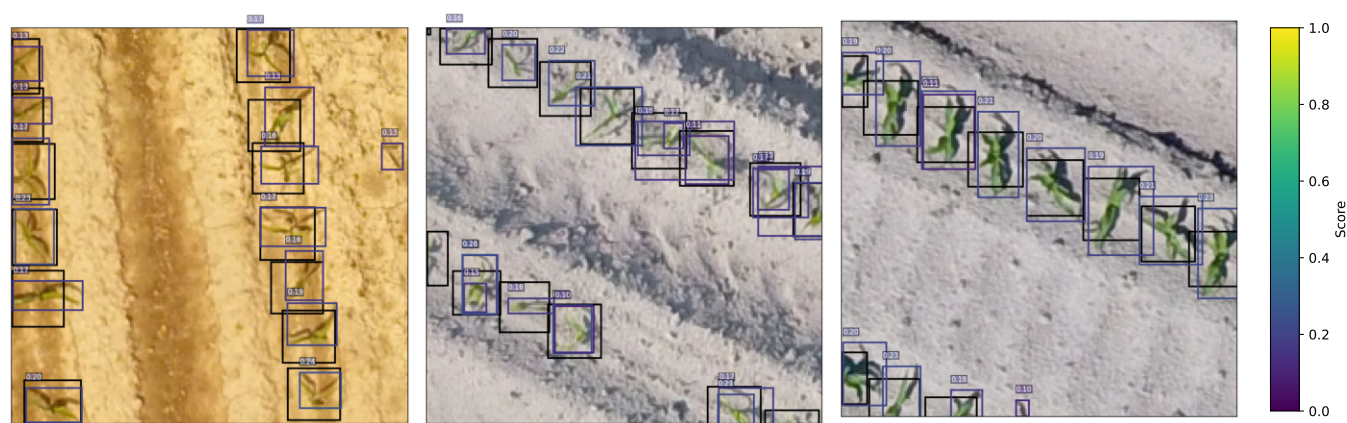


Figure 8. The best predictions with the OWLv2 model. The ID_1, ID_2 and ID_3 datasets respectively from the left hand side to the right. Prediction with owlv2-base-patch16-ensemble model of the ID_1 dataset, and with owlv2-base-patch16 model on the other two datasets. All the predictions are made with the prompt "seedling". Black bounding boxes are the ground truth annotations, while the bounding boxes in viridis color scale are the model predictions.

4. Discussion

The CD-ViTO model did not achieve the benchmark beacuse of the high variability of the errors as testify the high RMSE values. Still the best model was able to to predict the number of plants with a MAPE of 25%.

5. Conclusions

Author Contributions: For research articles with several authors, a short paragraph specifying their individual contributions must be provided. The following statements should be used “Conceptualization, X.X. and Y.Y.; methodology, X.X.; software, X.X.; validation, X.X., Y.Y. and Z.Z.; formal analysis, X.X.; investigation, X.X.; resources, X.X.; data curation, X.X.; writing—original draft preparation, X.X.; writing—review and editing, X.X.; visualization, X.X.; supervision, X.X.; project administration, X.X.; funding acquisition, Y.Y. All authors have read and agreed to the published version of the manuscript.”, please turn to the [CRediT taxonomy](#) for the term explanation. Authorship must be limited to those who have contributed substantially to the work reported.

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Data Availability Statement: We encourage all authors of articles published in MDPI journals to share their research data. In this section, please provide details regarding where data supporting reported results can be found, including links to publicly archived datasets analyzed or generated during the study. Where no new data were created, or where data is unavailable due to privacy or ethical restrictions, a statement is still required. Suggested Data Availability Statements are available in section “MDPI Research Data Policies” at <https://www.mdpi.com/ethics>.

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Abbreviations

The following abbreviations are used in this manuscript:

MDPI	Multidisciplinary Digital Publishing Institute
DOAJ	Directory of open access journals
TLA	Three letter acronym
LD	Linear dichroism

Appendix A

Appendix A.1

In summary, the HC algorithm is based on color thresholding, that rely on the manual setting of Hue, Saturation, and Value (HSV) color minimum and maximum thresholding. Thresholding creates a binary image that is subsequently regionalized by connected components. The connected components are then filtered by area. Centers of the areas are fitted with RANSAC to get the slope of the lines where plants lie. Only the tiles where the slope is within the interquartile range are considered for further processing. The centers of the filtered tiles are then fitted again with RANSAC to get the lines on the neighboring of the first line detected.

This as most of HC algorithm is intrinsically limited by the color thresholding. Our HC is based on the color of the maize seedlings, which is green.

Appendix B

All appendix sections must be cited in the main text. In the appendices, Figures, Tables, etc. should be labeled, starting with “A”—e.g., Figure A1, Figure A2, etc.

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Algorithm A1 H1

Require: *tiles* ▷ Orthomosaic tiles
Require: *col_range* ▷ Color space thresholds
Require: *leaf_area_range* ▷ Leaf area range in pixels
Ensure: *plants* ▷ List of polygons

```

1: function CONNECTED_COMPONENTS(binary_image) [87]
2:   return regions
3: for tile in tiles do
4:   mask ← {p ∈ tile | color(p) ∈ col_range}
5:   regions ← connected_components(mask)
6:   plants ← {region | region ∈ regions ∧ region.area ∈ leaf_area_range}
7: return plants

```

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Algorithm A2 H2

```

1: function PROCESS_TILES(intra – row_dist)
2:   observations  $\leftarrow \{\}$ 
3:   plants  $\leftarrow$  HC1(tiles)
4:   for tile in tiles do
5:     regions  $\leftarrow$  plants[tile]
6:     centroids  $\leftarrow$  REGION_CENTROIDS(regions)
7:     clust_cen  $\leftarrow$  AGGLOMERATE_REGIONS(regions, min_dist)
8:     inlier_points, model  $\leftarrow$  EXTRACT_RANSAC_LINE(clust_cen, intra – row_dist)
9:     line_length  $\leftarrow$  GET_LINE_LENGTH(model)
10:    expected_number_of_plants  $\leftarrow \frac{\text{line\_length}}{\text{intra\_row\_dist}}$ 
11:    if inlier_points  $\equiv$  expected_number_of_plants then
12:      observations[tile]  $\leftarrow$  (clust_cen, inlier_points, model)
13:  return observations
14: function FILTER_OBSERVATIONS_BY_SLOPE
15:   filtered_observations  $\leftarrow \{\}$ 
16:   for tile  $\in$  observations do
17:     slope  $\leftarrow$  observations[tile]['model']
18:     if model.slope  $\approx$  MEAN_SLOPE then
19:       filtered_observations[tile]  $\leftarrow$  observations[tile]
20:  return filtered_observations
21: function PROCESS_OBSERVATIONS(observations, inter – row_dist)
22:   objects  $\leftarrow \{\}$ 
23:   for tile  $\in$  observations do
24:     tile_centers  $\leftarrow$  observations[tile]['clust_cen']
25:     first_row_centers  $\leftarrow$  observations[tile]['inlier_points']
26:     first_row_model  $\leftarrow$  observations[tile]['model']
27:     centers  $\leftarrow \{p \mid p \in \text{tile\_centers} \wedge p \notin \text{first\_row\_centers}\}$ 
28:     second_row_centers, second_row_model  $\leftarrow$  EXTRACT_RANSAC_LINE(centers, intra –
      row_dist)
29:     line_length  $\leftarrow$  GET_LINE_LENGTH(second_row_model)
30:     expected_number_of_plants  $\leftarrow \frac{\text{line\_length}}{\text{intra\_row\_dist}}$ 
31:     if second_row_model.slope  $\approx$  first_row_model.slope then
32:       if  $\text{abs}(\text{second\_row\_model.intercept} - \text{first\_row\_model.intercept}) \approx \text{inter} -$ 
        row_dist then
33:         if second_row_centers  $\equiv$  expected_number_of_plants then
34:           objects[tile]  $\leftarrow$  (first_row_centers, second_row_centers)
35:  return objects
36: function MAIN
37:   observations  $\leftarrow$  PROCESS_TILES(intra – row_dist)
38:   MEAN_SLOPE  $\leftarrow$  MEAN(observations['model'])
39:   observations  $\leftarrow$  FILTER_OBSERVATIONS_BY_SLOPE(observations, MEAN_SLOPE)
40:   objects  $\leftarrow$  PROCESS_OBSERVATIONS(observations, inter – row_dist)
41:  return objects

```

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